

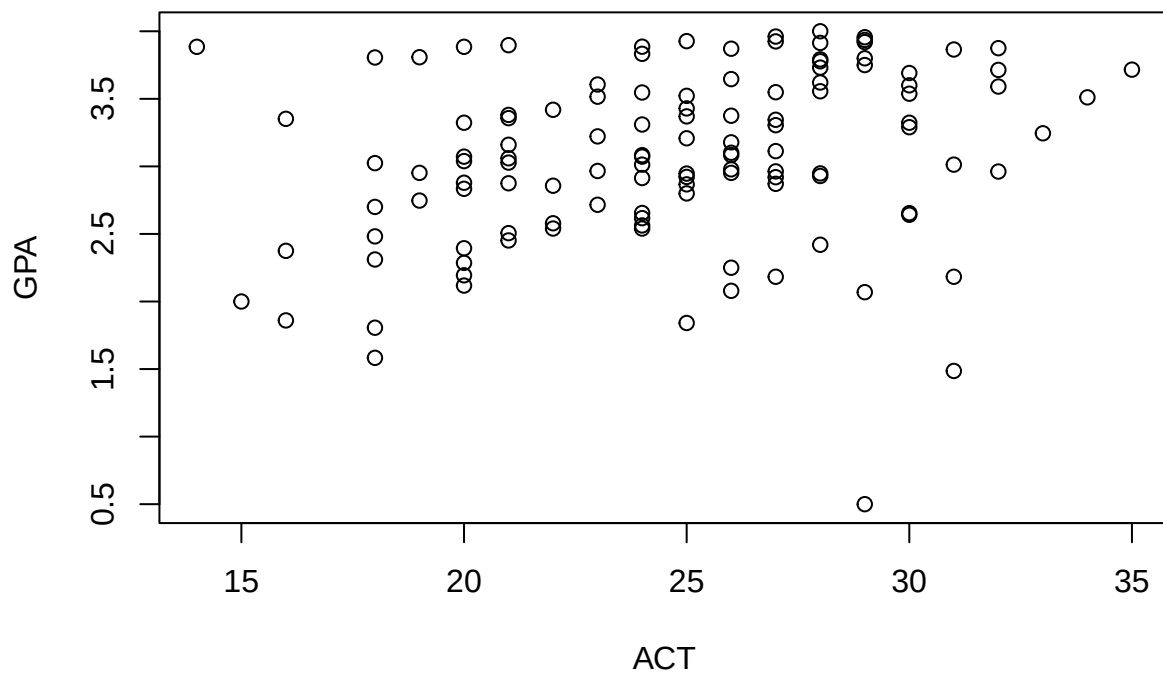
417-HW1

Alex Caramanico

October 8, 2022

Problem 1

```
GPA = read.csv("C:/Users/jacma/Downloads/GPA Dataset.csv") # load dataset
# a
ACT = GPA[,2] # ACT scores in second column
GPA = GPA[,3] # College GPA in third column
plot(ACT,GPA)
```



```
# The plot seems to have a slight upward trend, but there also appears to be a
# large number of datapoints which deviate from said trend (e.g. the student
# with a GPA of 0.5).
```

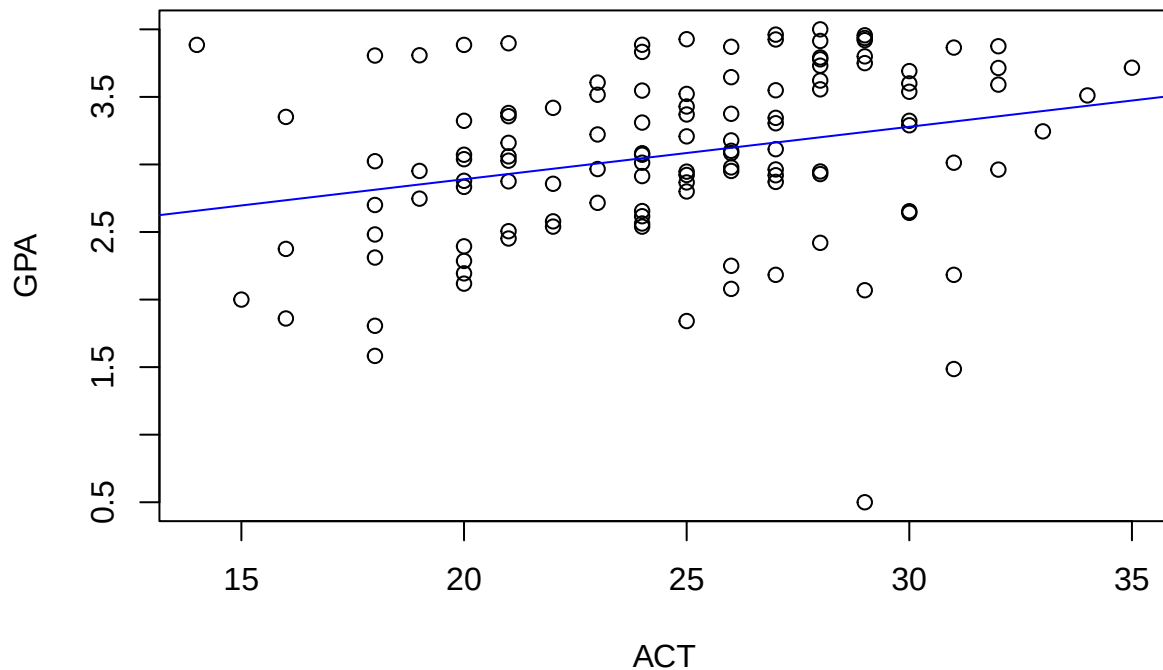
```

# b
fit = lm(GPA~ACT) #  $y = B_0 + B_1 \cdot ACT + e$ 
summary(fit) # estimated function:  $\hat{y} = 2.11405 + 0.03883X$ 

##
## Call:
## lm(formula = GPA ~ ACT)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.74004 -0.33827  0.04062  0.44064  1.22737
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.11405     0.32089   6.588 1.3e-09 ***
## ACT          0.03883     0.01277   3.040 0.00292 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6231 on 118 degrees of freedom
## Multiple R-squared:  0.07262,    Adjusted R-squared:  0.06476
## F-statistic: 9.24 on 1 and 118 DF,  p-value: 0.002917

# c
yhat = predict(fit)
plot(ACT,GPA)
abline(fit, col='blue')

```

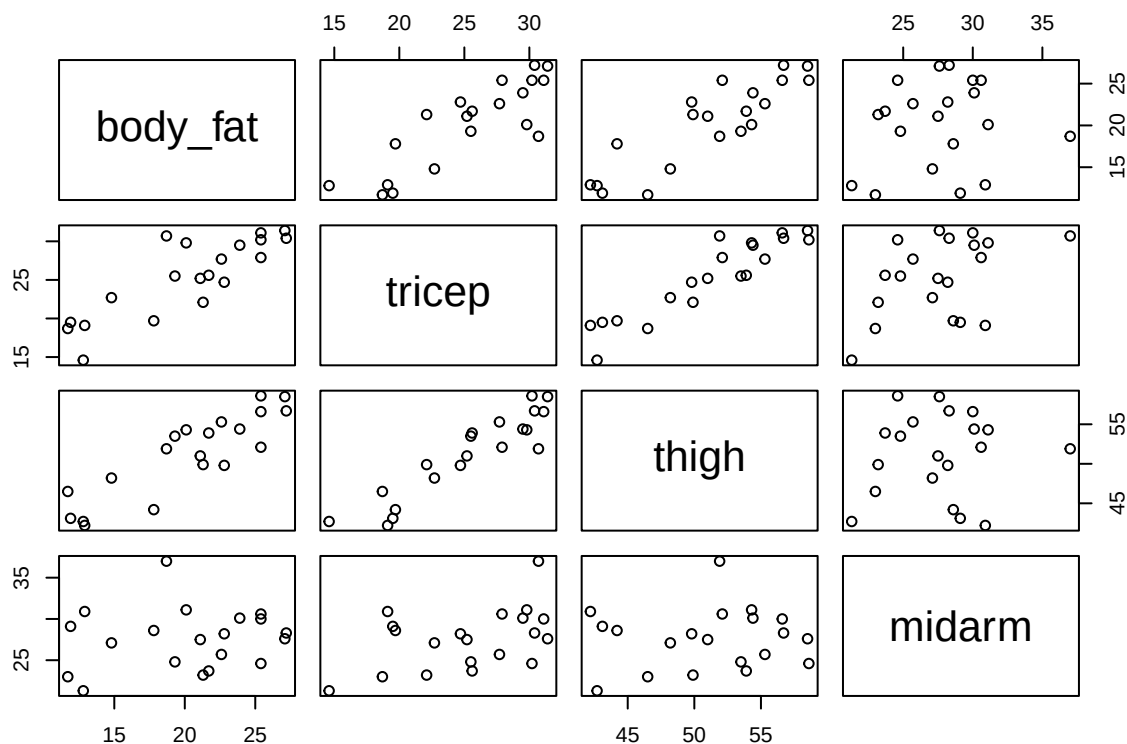


Estimated function does not appear to fit the data particularly well, there are a great deal of large residuals (where data points are far from the regression line), which leads to a greater sum of squared errors. The R^2 of 0.07262 means that only ~7.2% of the variance in college GPA is explained by ACT scores. Perhaps there are other variables which could better predict college GPA if included in a regression. It is worth noting that despite the low R^2 , ACT scores are significant at the 99% confidence level. This is probably because it is the only explanatory variable in the regression.

Problem 2

```
BF = read.csv("C:/Users/jacma/Downloads/BodyFat Dataset.csv") # load dataset

# a
tricep = BF[,2] # X1
thigh = BF[,3] # X2
midarm = BF[,4] # X3
body_fat = BF[,5] # Y
refined_data = cbind(body_fat, tricep, thigh, midarm) # data with needed variables only
pairs(refined_data)
```



```
X_i = cbind(tricep,thigh,midarm) # combine columns of explanatory variables
cor(X_i) # correlation matrix
```

```
##           tricep      thigh      midarm
## tricep 1.0000000 0.9238425 0.4577772
## thigh  0.9238425 1.0000000 0.0846675
## midarm 0.4577772 0.0846675 1.0000000
```

```
model = lm(body_fat~tricep+midarm+thigh)
library(car) # library to run vif() function
```

```
## Loading required package: carData
```

```
vif(model) # Variance inflation factors are extremely high for all three
```

```
##      tricep      midarm      thigh
## 708.8429 104.6060 564.3434
```

```
# explanatory variables, suggesting there is multicollinearity between them
```

```
# Interpretations
```

```
# Based on the scatter plot, variables tricep and thigh appear to be highly
# positively correlated, as they increase in tandem. In addition, tricep and
```

```
# midarm appear to have a slight positive correlation, also increasing together.  
# Thigh and midarm appear to be less correlated according to the scatter plot.
```

```
# The above observations are supported by the correlation matrix, which shows  
# the correlation between tricep and thigh to be 0.92, a very strong positive  
# relationship. This suggests multicollinearity, which means that their  
# respective observations exhibit very similar trends. This could cause issues  
# because each variable will likely diminish the significance of the other,  
# possibly making the other variable redundant in a linear regression model.
```

```
# As for the apparent relationship between each explanatory variable and the  
# dependent variable, both thigh and tricep seem to be strongly positively  
# correlated with body fat (which is to be expected due to the high correlation  
# between said independent variables), whereas midarm seems to have a more  
# random relationship (i.e. much less correlated).
```

```
# b  
reg = lm(body_fat~tricep+thigh+midarm) #  $y = B_0 + B_1X_1 + B_2X_2 + B_3X_3 + e$   
summary(reg) #  $\hat{y} = 117.085 + 4.334X_1 - 2.857X_2 - 2.186X_3$ 
```

```
##  
## Call:  
## lm(formula = body_fat ~ tricep + thigh + midarm)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -3.7263 -1.6111  0.3923  1.4656  4.1277   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)  117.085     99.782   1.173   0.258      
## tricep        4.334       3.016   1.437   0.170      
## thigh       -2.857       2.582  -1.106   0.285      
## midarm       -2.186       1.595  -1.370   0.190      
##  
## Residual standard error: 2.48 on 16 degrees of freedom  
## Multiple R-squared:  0.8014, Adjusted R-squared:  0.7641   
## F-statistic: 21.52 on 3 and 16 DF,  p-value: 7.343e-06
```

```
# The model appears to fit the data relatively well with an  $R^2$  of  $\sim 0.8$ , meaning  
# 80% of the variance in body fat is explained by independent variables. However,  
# none of the explanatory variables are significant at even the 90% confidence  
# level, which is likely a result of the multicollinearity discussed previously.  
# This is to say that although the model "fits" the data decently, there is  
# probably a better model which focuses only on the most significant variables.
```

```
# It should be noted that the adjusted  $R^2$  value is  $\sim 0.76$ , about 4 percentage  
# points lower than the normal  $R^2$ , further suggesting at least one of the  
# explanatory variables is superfluous.
```

```
# c  
#  $H_0: B_0 = B_1 = B_2 = B_3 = 0$  vs.  $H_1: \text{At least one inequality, } \alpha = 0.05$ 
```

```
# Decision rule: Reject null hypothesis if p-value < 0.05
# There is certainly a significant global regression at 5% level of significance.
# The p-value from the output is 0.000007343, which is far less than 0.05. Thus,
# the conclusion is that at least one of the independent variables is significant
# in explaining the variance of the body fat data. In other words, the test
# implies that one or more of B1, B2, or B3 is not equal to 0. The extremely
# small p-value suggests that at least one of the variables is almost certainly
# significant. This can be said with very high confidence (~99.99999%), meaning
# that only a minute number of potential samples would find results where none
# of the explanatory variables are significant (about 1 in 10 million)
```

```
# d
library(olsrr) # "import" model selection functions
```

```
##
## Attaching package: 'olsrr'

## The following object is masked from 'package:datasets':
##
## rivers
```

```
ols_step_all_possible(reg) # perform step-wise selection
```

```
##      Index N      Predictors R-Square Adj. R-Square Mallow's Cp
## 2      1 1          thigh 0.7710414    0.75832149    2.441959
## 1      2 1          tricep 0.7110966    0.69504643    7.270340
## 3      3 1          midarm 0.0202903   -0.03413801   62.912811
## 5      4 2      tricep midarm 0.7861598    0.76100218    3.224212
## 4      5 2      tricep thigh 0.7780518    0.75194029    3.877289
## 6      6 2      thigh midarm 0.7757123    0.74932549    4.065734
## 7      7 3 tricep thigh midarm 0.8013586    0.76411328    4.000000
```

```
reg2 = lm(body_fat~tricep+midarm) # updated regression equation
summary(reg2) # both explanatory variables significant at alpha = 0.05
```

```
##
## Call:
## lm(formula = body_fat ~ tricep + midarm)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.8794 -1.9627  0.3811  1.2688  3.8942
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   6.7916     4.4883   1.513  0.1486
## tricep         1.0006     0.1282   7.803 5.12e-07 ***
## midarm        -0.4314     0.1766  -2.443  0.0258 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 2.496 on 17 degrees of freedom
## Multiple R-squared:  0.7862, Adjusted R-squared:  0.761
## F-statistic: 31.25 on 2 and 17 DF,  p-value: 2.022e-06
```

```
vif(reg2) # re-run vif function, inflation factors for both variables are low
```

```
##      tricep      midarm
## 1.265118 1.265118
```

```
# The adjusted R^2 value decreased slightly, but since all variables are now
# significant, reg2 is arguably a better model as the relationship between
# body fat and tricep thickness and midarm circumference is more statistically
# significant. Thus, the adjusted model is likely more useful in making
# predictions than the full model.
```

```
# Additionally, this model had the second lowest Cp score according to the
# dataframe output, second only to the model which only included thigh data.
# reg2 has both a higher R^2 and adjusted R^2 than this model, and thus is more
# useful and explanatory than (any of) the true effect model(s).
```

```
# Although tricep and thigh were highly positively correlated, replacing tricep
# with thigh in a regression made midarm highly insignificant (p-value = 0.5).
# Additionally, the adjusted R^2 for the thigh + midarm model was lower
# Putting all together, the final model is  $y = B_0 + B_1X_1 + B_2X_3 + e$ ,
# where  $X_1$  is tricep thickness and  $X_3$  is midarm circumference.
# The final estimated equation is  $\hat{y} = 6.7961 + 1.0006X_1 - 0.4314X_3$ 
```

```
# In all, reg2 was chosen as it is very balanced with desirable R^2 and Cp
# values in addition to all of its parameters being significant at  $\alpha = 0.05$ 
```